CCTV Engineering Report

Technical Engineering Report

DRAFT - For engineering review; site verification required

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1. Project Scope, Objectives, and Design Basis

This report defines the engineering basis for a closed-circuit television (CCTV) system serving the project site identified as TBD. The scope includes development of the system concept, camera coverage requirements, equipment and network interfaces, power and recording provisions, cybersecurity considerations, installation requirements, commissioning activities, and handover documentation. The final number of cameras, exact locations, viewing directions, retention period, recording resolution, and other project-specific values remain TBD until site information and stakeholder requirements are confirmed.

The primary objective is to provide a reliable and maintainable video surveillance system that supports security monitoring, incident review, operational awareness, and, where applicable, investigation of access, safety, or asset-related events. The system should provide appropriately selected views of designated entrances, circulation routes, restricted areas, critical assets, loading or service zones, and other locations identified by the operator. Coverage shall be designed to meet the intended viewing objective for each camera, such as detection, observation, recognition, or identification. These objectives shall be documented in the camera schedule and verified during commissioning.

The design basis is a networked, IP-based CCTV system using fixed or movable cameras as required by the approved coverage plan. Cameras are expected to connect through Power over Ethernet (PoE) network switches, with video transported to a recording and management platform. The proposed architecture shall support centralized monitoring, authorized playback, event export, user administration, time synchronization, system health monitoring, and controlled integration with relevant security or building systems where required. The final manufacturers, models, software versions, licensing requirements, and interoperability constraints are TBD and shall be confirmed before procurement.

The design shall prioritize useful image quality, suitable field of view, adequate lighting performance, environmental compatibility, secure communications, and maintainability. Camera selection shall consider scene geometry, mounting height, backlighting, night conditions, illumination levels, weather exposure, vandal resistance, and required identification distance. Any reliance on infrared illumination, supplementary lighting, analytics, audio, edge storage, or pan-tilt-zoom functionality shall be explicitly identified and approved.

All performance criteria that depend on the site, client standards, applicable regulations, or operational policy shall be confirmed during detailed design. These include coverage boundaries, privacy masking, retention duration, frame rate, image resolution, acceptable availability, network segregation, access permissions, export format, and recovery requirements. No test, inspection, coverage result, or acceptance status is assumed complete unless recorded in the commissioning documentation. Values not established by the project team shall remain marked TBD rather than being inferred.

2. Site Conditions, Coverage Requirements, and Assumptions

The CCTV system shall be designed to suit the final site layout, operational requirements, environmental conditions, and security risk profile. At the time of this report, detailed information regarding the site address, building configuration, floor plans, perimeter boundaries, lighting conditions, existing infrastructure, and operating hours is not available. These items shall be confirmed through an approved site survey and coordinated design review before equipment selection and installation. Any dimensions, cable routes, mounting heights, equipment-room locations, network characteristics, and power availability not identified in the project documentation shall remain TBD.

Coverage shall be developed from defined viewing objectives rather than camera quantity alone. The design should consider entrances and exits, vehicle and pedestrian access points, reception and public areas, circulation routes, loading or service areas, critical rooms, restricted zones, storage areas, plant spaces, and the site perimeter where applicable. Each proposed camera location shall have a documented purpose, including general

monitoring, activity detection, recognition, identification, or evidential recording. Camera fields of view shall be coordinated to minimize blind spots while avoiding unnecessary duplication, excessive glare, and views into areas where privacy restrictions apply.

The final coverage assessment shall account for daylight, low-light, artificial-light, backlighting, shadows, weather exposure, dust, moisture, vibration, and potential obstruction by vehicles, vegetation, temporary works, or future construction. Where image quality depends on supplementary illumination, the requirement for existing or additional lighting shall be identified and coordinated with the electrical and security designs. Outdoor cameras and associated accessories shall be selected for the actual environmental exposure and shall include suitable protection against water, dust, corrosion, temperature variation, and tampering, as applicable to the confirmed location.

It is assumed that the client will provide current architectural, electrical, network, and security information where available, together with access to the site for surveys and verification. It is also assumed that camera mounting surfaces can support the proposed equipment and that suitable routes can be established for structured cabling, containment, and power distribution. These assumptions require confirmation; no structural capacity, cable distance, network bandwidth, PoE availability, or recording duration shall be considered verified without documented review.

The design shall comply with applicable legislation, codes, client standards, privacy requirements, and site-specific safety procedures. Camera placement, signage, access to recorded material, retention periods, and monitoring responsibilities shall be agreed with the client and relevant stakeholders. Areas excluded from surveillance shall be identified where required. Final coverage, equipment positions, and performance requirements shall be validated during design review, installation inspection, and commissioning, with unresolved items recorded as TBD or included in the project’s outstanding information register.

3. CCTV System Architecture and Equipment Schedule

The CCTV system shall be designed as a networked, IP-based surveillance system comprising fixed and/or adjustable network cameras, Power over Ethernet (PoE) switching, network connectivity, recording and management servers or network video recorders, operator workstations, displays, storage, and associated mounting and containment infrastructure. The final architecture shall be confirmed against the approved coverage requirements, site conditions, network availability, cybersecurity policy, and operational monitoring requirements. Any equipment quantity, model, performance rating, or location not established by the project information shall remain identified as TBD until design review and site verification are complete.

Cameras shall be selected according to the viewing objective at each location, such as general scene observation, entry and exit monitoring, vehicle identification, perimeter surveillance, or identification of persons and activities. Camera types may include fixed dome, fixed bullet, box, panoramic, or pan-tilt-zoom units where justified by the application. Each camera shall provide a suitable image sensor, lens or focal-length range, resolution, low-light performance, infrared or other supplementary illumination where required, and environmental protection appropriate to its installation location. The final camera schedule shall record the camera identifier, location, viewing objective, lens and resolution, PoE requirement, and verification status.

Camera traffic shall be transported over the site data network using dedicated or logically segregated CCTV network infrastructure. PoE switches shall provide sufficient port capacity, power budget, uplink bandwidth, and environmental suitability for the connected devices. Switches, fibre or copper uplinks, patching, cabinets, and network outlets shall be coordinated with the wider IT and building services design. The required network topology, transmission media, switch locations, and resilience arrangements are TBD unless confirmed in the approved design documentation.

Recording shall be provided by a network video recorder, video management system, or equivalent server-based platform. The recording platform shall support the specified camera types, continuous and event-based recording, time synchronisation, user access control, audit logging, export of evidential footage, health monitoring, and retention requirements. Recording duration, image frame rate, storage capacity, redundancy, and archive arrangements shall be calculated from the approved camera count and operational requirements; these values are TBD where not yet defined.

The equipment schedule shall include cameras, mounting brackets, junction boxes, PoE switches, network cabinets, patch panels, fibre or copper links, recording equipment, storage, management workstations, operator displays, uninterruptible power supplies, surge protection, and required software licences. Equipment shall be compatible, supportable, and installed in accordance with manufacturer requirements. Final selections shall be submitted for approval, including datasheets, environmental ratings, power requirements, licence dependencies, and proposed equipment quantities.

4. Camera Schedule Template: Camera ID, Location, Viewing Objective, Lens/Resolution, PoE, and Verification Status

The camera schedule shall be the controlling register for identifying each CCTV device, its intended field of view, technical configuration, power requirement, and verification status. Camera IDs should remain unique throughout design, installation, configuration, testing, drawings, and handover documentation. Locations should reference the approved site plan, room name, elevation, or access point once confirmed. Where the project information is incomplete, the entry shall be retained as TBD rather than supplemented by an assumed value.

The viewing objective should describe the operational purpose of the camera, such as monitoring an entrance, documenting visitor approach, observing a circulation route, protecting an asset, or providing general situational awareness. The objective should be sufficiently specific to support later image-quality checks and camera positioning decisions. Final lens selection and resolution shall be based on the required field of view, target detail, lighting conditions, mounting position, and applicable identification or recognition criteria. These values remain TBD until the design basis and equipment selection are approved.

PoE information should identify the intended power and data interface, including the applicable PoE standard or switch-port allocation where available. The schedule should be coordinated with the network and power design so that switch capacity, port availability, cable routes, and any device power budget can be confirmed. Verification status shall distinguish between design review, installation inspection, configuration, functional testing, and final acceptance. No camera should be recorded as verified unless the relevant inspection or test has been completed and documented.

Camera ID	Location	Viewing Objective	Lens/Resolution	PoE	Verification Status
CAM-001	TBD	TBD	TBD	TBD	Pending verification
CAM-002	TBD	TBD	TBD	TBD	Pending verification
CAM-003	TBD	TBD	TBD	TBD	Pending verification
CAM-004	TBD	TBD	TBD	TBD	Pending verification
CAM-005	TBD	TBD	TBD	TBD	Pending verification
CAM-006	TBD	TBD	TBD	TBD	Pending verification

The completed schedule should be reviewed against architectural drawings, security objectives, privacy constraints, lighting conditions, and the equipment schedule before procurement or installation. Any change to a

camera’s position, coverage objective, lens, resolution, or power arrangement should be recorded through the project change-control process. The final handover version should include the approved camera identifier, as-installed location, configuration details, associated network port where applicable, and the documented verification outcome. Unresolved entries shall remain clearly marked TBD and be transferred to the outstanding-information register for closure.

5. Network, Power, Recording, Storage, and Cybersecurity Requirements

The CCTV system shall use a dedicated or logically segregated video network, subject to the approved information technology architecture. Camera traffic shall be carried on an assigned VLAN or equivalent security boundary, with routing to management, recording, and monitoring systems restricted to approved services. Network switches shall provide sufficient PoE capacity for the connected cameras, including allowance for startup demand, infrared illumination, heater or blower loads where applicable, and reasonable future expansion. PoE standard, switch port allocation, uplink capacity, addressing scheme, and network equipment locations shall be confirmed during detailed design and recorded as TBD where not yet provided.

All cameras, switches, recording servers, storage systems, and associated active equipment shall be connected to appropriately rated power supplies. Critical CCTV equipment shall be supported by an uninterruptible power supply where required by the operational risk assessment or client standards. UPS autonomy, standby generation interface, distribution-board source, earthing arrangements, surge protection, and power monitoring requirements remain project-specific and shall be confirmed before procurement. Power and data cabling shall be routed and protected to reduce exposure to mechanical damage, electromagnetic interference, water ingress, and unauthorized access.

Recording shall be performed by the approved video management system or network video recorder. Recording profiles shall be configured according to the approved camera schedule and viewing objectives, including resolution, frame rate, compression, motion or event rules, and continuous recording requirements. The required retention period, recording availability target, evidential export format, and maximum acceptable loss of recorded video following a system or power failure are TBD unless defined by the project brief. Storage capacity shall be calculated from the final camera count, stream parameters, recording mode, retention period, redundancy method, metadata requirements, and reserve capacity. Any assumptions used in the calculation shall be documented and approved.

The recording platform shall provide controlled user access, audit logging, time synchronization, health monitoring, alarm reporting, and protected export of video and associated metadata. System clocks shall use an approved time source so that events from cameras, access control, network equipment, and other systems can be correlated. Storage shall be protected against unauthorized alteration and configured with appropriate redundancy or backup arrangements where required.

Cybersecurity controls shall include unique administrative credentials, role-based access, strong authentication where supported, secure management protocols, timely firmware and software maintenance, disabled unused services, and controlled remote access. Default passwords shall be changed before connection to the operational network. Cameras and servers shall not be directly exposed to the public internet. Firewall rules, antivirus or endpoint controls where applicable, vulnerability-management responsibilities, backup procedures, and incident-response contacts shall be agreed with the system owner and IT authority. Configuration files, passwords, encryption keys, and exported footage shall be handled as controlled information.

6. Installation, Configuration, and Handover Requirements

Installation shall be carried out by suitably qualified personnel in accordance with the approved design, manufacturer instructions, applicable electrical and communications standards, and site health and safety procedures. Camera locations, mounting arrangements, fields of view, cable routes, equipment-room interfaces, and network connection points shall be verified against the approved drawings before permanent installation. Any required deviation shall be recorded and submitted for review before implementation. Final camera positions shall maintain the intended viewing objective while avoiding unacceptable obstruction, glare, excessive backlighting, and privacy impacts.

Cameras, network equipment, recording appliances, displays, storage systems, and ancillary components shall be securely mounted, labelled, and protected against environmental conditions appropriate to their installation locations. Cabling shall be installed with suitable segregation from sources of electrical interference, supported throughout its route, and terminated in accordance with the approved structured-cabling requirements. Outdoor and exposed installations shall include appropriate weatherproofing, sealing, drainage, and corrosion protection. Unused ports, entries, and openings shall be secured against dust, moisture, and unauthorised access.

Configuration shall include unique device names and addresses, time synchronisation, approved video profiles, image orientation, privacy masking where required, motion or analytics settings where specified, recording schedules, retention parameters, user roles, and alarm or event actions. Default credentials shall be changed before operational use. Administrative access shall be restricted to authorised personnel, with strong authentication and least-privilege permissions applied where supported. Network, firewall, remote-access, and cybersecurity settings shall conform to the project requirements. All configuration values not defined by the approved design shall remain TBD until formally confirmed.

Before handover, each camera and associated system function shall be inspected and tested using the commissioning checklist. Verification shall include image availability, identification and labelling, focus and field of view, day/night operation where applicable, infrared performance where applicable, PoE connectivity, recording and playback, time accuracy, storage operation, user access, event handling, and loss-of-connection recovery. Actual results shall be recorded; no item shall be marked complete without documented evidence. Defects shall be entered in a corrective-action register and closed or formally accepted as outstanding.

The handover package shall include as-installed drawings, the final camera schedule, equipment and serial-number records, cable and port schedules, approved configuration backups, IP and credential-management records held through the approved secure process, operating and maintenance instructions, warranty information, licences, test records, defect status, and training records. The installer shall provide operator and administrator familiarisation covering routine operation, playback, export, alarm response, fault reporting, and basic maintenance. Acceptance shall require review of the handover documents and sign-off by the authorised project representatives.

7. Commissioning Checklist Template: Item, Test Method, Expected Result, Actual Result, and Status

Commissioning shall verify that each CCTV component is installed, configured, connected, and operating in accordance with the approved design, manufacturer requirements, and project acceptance criteria. The checklist below is a controlled template. The Actual Result and Status fields shall be completed during witnessed testing; no item shall be recorded as passed without objective evidence or an approved reason for deviation. Status values may include Pass, Fail, Pending, Not Applicable, or TBD, as appropriate.

Item	Test Method	Expected Result	Actual Result	Status
Camera identification and location	Compare installed camera labels and locations with the approved camera schedule and drawings.	Camera ID, location, orientation, and asset label correspond with the approved documentation.	TBD	TBD
Physical installation	Inspect mounting, housing, fixings, seals, cable entry, and access for maintenance.	Camera is secure, correctly aligned, weather-protected where required, and free from visible damage.	TBD	TBD
Field of view and image quality	View live images from the intended monitoring position under representative lighting conditions.	Required scene is visible, focus is acceptable, image is stable, and obstructions or excessive glare are not present.	TBD	TBD
Camera configuration	Review resolution, frame rate, encoding, time settings, profiles, and user permissions against approved settings.	Configuration matches the design basis and approved cybersecurity requirements.	TBD	TBD
Day/night and illumination performance	Test under available daytime, low-light, and infrared conditions, where applicable.	Camera changes modes correctly and provides usable images for the defined viewing objective.	TBD	TBD
PoE and network connectivity	Confirm switch port assignment, link status, VLAN allocation, and power negotiation using approved network tools.	Camera receives the required power, communicates on the designated network, and remains stable.	TBD	TBD
Video management system integration	Add or verify the camera in the VMS and review live view, recording, playback, and alarm functions.	Camera is correctly identified and provides continuous live and recorded video in the VMS.	TBD	TBD
Recording and retention	Review recording schedules and, where permitted, perform a playback test across the configured recording modes.	Recording occurs according to the approved schedule and retention requirement; playback is available.	TBD	TBD
Time synchronisation	Compare camera, VMS, server, and reference time sources during the test.	System clocks are synchronised within the project-approved tolerance.	TBD	TBD
Alarm and event handling	Trigger representative analytics, input, tamper, loss-of-video, or other configured events.	Events are generated, displayed, logged, and escalated to the designated recipients as designed.	TBD	TBD

Item	Test Method	Expected Result	Actual Result	Status
Cybersecurity and access control	Review accounts, passwords, firmware status, services, remote access, and audit logging.	Unauthorised access is restricted, required logs are enabled, and approved hardening measures are applied.	TBD	TBD
Failover and recovery	Simulate an agreed power, network, or recording-service interruption where safe to do so.	System recovers in accordance with the approved resilience and recovery requirements.	TBD	TBD
Documentation and training	Review as-built drawings, configuration backups, test records, manuals, and operator training evidence.	Handover information is complete, current, and accepted by the responsible parties.	TBD	TBD

Any failed or incomplete item shall be assigned a reference number, responsible party, corrective action, and target completion date. Retesting shall be recorded against the original checklist item, with supporting evidence such as screenshots, configuration exports, test recordings, or signed inspection records retained in the commissioning file. Final acceptance shall remain pending until all critical defects are closed or formally accepted as approved deviations.

8. Outstanding Information, TBD Values, Limitations, and Acceptance Sign-Off

The following information remains outstanding and shall be confirmed before final design approval, installation completion, or operational acceptance: approved camera locations and mounting details; final viewing objectives and privacy boundaries; camera models, lens selections, resolutions, and firmware versions; PoE switch capacity and port allocation; network addressing, VLAN, routing, and firewall requirements; recording platform configuration; retention duration; storage capacity; time-synchronisation source; UPS autonomy; equipment-room environmental conditions; and the final cable routes and containment requirements. Any value not confirmed in the approved design documentation shall remain identified as TBD and shall not be treated as an implicit acceptance of the design.

Site-dependent matters also require verification during installation. These include line of sight, image obstruction, lighting conditions, glare, backlighting, focus, field of view, mounting strength, access for maintenance, and compliance with applicable privacy, safety, and security requirements. Changes resulting from site conditions shall be recorded through the project change-control process. A revised camera schedule, as-built drawing set, network information, and equipment schedule shall be issued where installed conditions differ from the approved design.

This report is subject to the limitations of the information available at the time of preparation. No unverified measurements, coverage performance, network capacity, storage calculation, cybersecurity control, or commissioning result is represented as complete. The templates in this report are planning and verification tools; blank, TBD, or unverified entries require completion by the responsible project parties. Acceptance shall be based on documented inspection and test evidence, not solely on equipment delivery, visual installation quality, or nominal manufacturer performance.

Before handover, the installer shall provide completed commissioning records, equipment data sheets, configuration backups where permitted, test results, defect and close-out records, warranty information, training records, and current as-built documentation. The operator or client representative shall review the delivered system against the approved requirements and identify any exclusions, residual defects, or operational constraints. Open

items shall include an owner, target completion date, and agreed disposition.

Acceptance sign-off shall be completed only after all mandatory commissioning items have a recorded actual result and status, all critical defects have been closed or formally accepted, and the required handover documentation has been received. Signatories confirm review of the delivered CCTV system within the stated scope; they do not waive statutory obligations, approved performance requirements, or unresolved risks that are expressly recorded below.

Client/Owner representative: TBD Name and role: TBD Signature: TBD Date: TBD

Installer/Integrator representative: TBD Name and role: TBD Signature: TBD Date: TBD

Consultant/Engineer representative: TBD Name and role: TBD Signature: TBD Date: TBD